EXPERIMENT NO. 3

AIM-To create an Employee-Department-Project schema and perform SQL queries using Selection, Projection, Aggregate Functions, GROUP BY, HAVING, CASE and ORDER BY.

- CREATE TABLE-

queries.sql

```sql
CREATE TABLE Department(
 DeptID INT PRIMARY KEY,
 DeptName VARCHAR(30)
);

CREATE TABLE Project(
 ProjectID INT PRIMARY KEY,
 ProjectName VARCHAR(50),
 Budget INT,
 DeptID INT
);

CREATE TABLE Employee(
 EmpID INT PRIMARY KEY,
 EmpName VARCHAR(30),
 Salary INT,
 DeptID INT,
 ProjectID INT
);
```

- INSERT DATA-

```sql
INSERT INTO Department VALUES
(1,'IT'),(2,'HR'),(3,'Finance'),(4,'Marketing'),(5,'Operations');

INSERT INTO Project VALUES
(101,'Cloud Migration',800000,1),
(102,'AI Chatbot',1200000,1),
(103,'Employee Portal',500000,2),
(104,'Payroll System',700000,3),
(105,'Market Analysis',600000,4),
(106,'Digital Campaign',900000,4),
(107,'Supply Chain',1000000,5),
(108,'Inventory Management',750000,5);
```

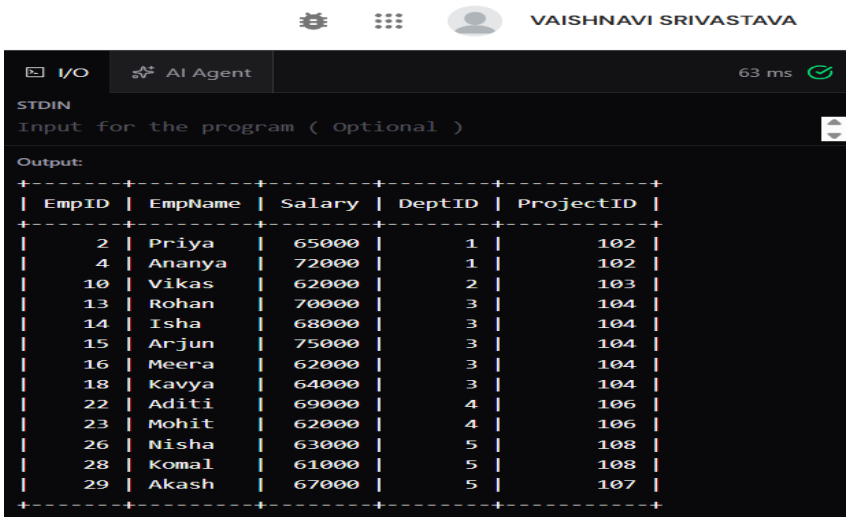
- INSERT 30 EMPLOYEES DATA-

```
INSERT INTO Employee VALUES
(1,'Aarav',55000,1,101),
(2,'Priya',65000,1,102),
(3,'Rahul',48000,1,101),
(4,'Ananya',72000,1,102),
(5,'Karan',60000,1,101),
(6,'Neha',58000,1,102),
(7,'Riya',45000,2,103),
(8,'Aditya',52000,2,103),
(9,'Simran',48000,2,103),
(10,'Vikas',62000,2,103),
(11,'Pooja',55000,2,103),
(12,'Manish',50000,2,103),
(13,'Rohan',70000,3,104),
(14,'Isha',68000,3,104),
(15,'Arjun',75000,3,104),
(16,'Meera',62000,3,104),
(17,'Nitin',58000,3,104),
(18,'Kavya',64000,3,104),
(19,'Sahil',54000,4,105),
(20,'Sneha',60000,4,106),
(21,'Varun',57000,4,105),
(22,'Aditi',69000,4,106),
(23,'Mohit',62000,4,106),
(24,'Tanya',56000,4,105),
(25,'Dev',59000,5,107),
(26,'Nisha',63000,5,108),
(27,'Yash',55000,5,107),
(28,'Komal',61000,5,108),
(29,'Akash',67000,5,107),
(30,'Divya',58000,5,108);
```

- REQUIRED QUERIES

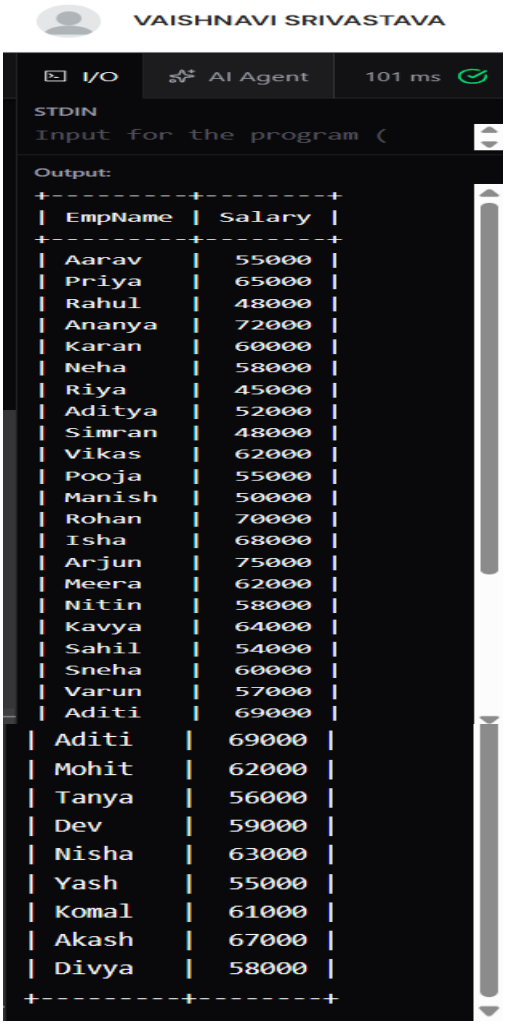
1. SELECTION-

---- SELECT * FROM Employee
WHERE Salary > 60000;



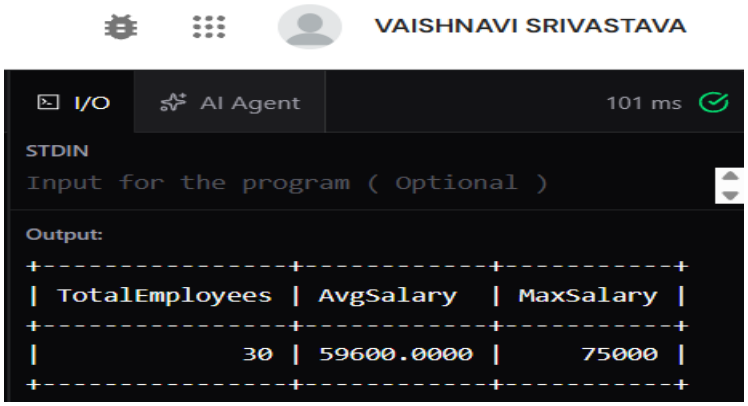
2. PROJECTION-

---- SELECT EmpName, Salary FROM Employee;



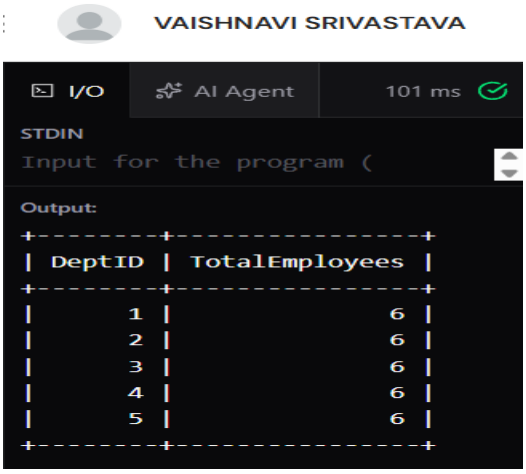
3. AGGREGATE-

---- SELECT COUNT(*) AS TotalEmployees,
AVG(Salary) AS AvgSalary,
MAX(Salary) AS MaxSalary
FROM Employee;



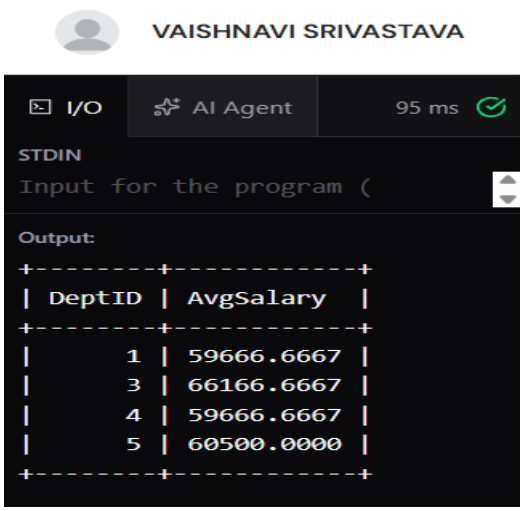
4. GROUP BY-

---- SELECT DeptID, COUNT(*) AS TotalEmployees
FROM Employee
GROUP BY DeptID;



5. HAVING BY-

---- SELECT DeptID, AVG(Salary) AS AvgSalary
FROM Employee
GROUP BY DeptID
HAVING AVG(Salary) > 55000;



6. CASE -

---- SELECT EmpName, Salary,
CASE
WHEN Salary >= 70000 THEN 'High'
WHEN Salary >= 60000 THEN 'Medium'
ELSE 'Low'
END AS SalaryLevel
FROM Employee;

VAISHNAVI SRIVASTAVA

I/O | AI Agent | 85 ms

STDIN

Input for the program (Optional)

Output:

```
+---------+--------+-------------+
| EmpName | Salary | SalaryLevel |
+---------+--------+-------------+
| Aarav   |  55000 | Low         |
| Priya   |  65000 | Medium      |
| Rahul   |  48000 | Low         |
| Ananya  |  72000 | High        |
| Karan   |  60000 | Medium      |
| Neha    |  58000 | Low         |
| Riya    |  45000 | Low         |
| Aditya  |  52000 | Low         |
| Simran  |  48000 | Low         |
| Vikas   |  62000 | Medium      |
| Pooja   |  55000 | Low         |
| Manish  |  50000 | Low         |
| Rohan   |  70000 | High        |
| Isha    |  68000 | Medium      |
| Arjun   |  75000 | High        |
| Meera   |  62000 | Medium      |
| Nitin   |  58000 | Low         |
| Kavya   |  64000 | Medium      |
| Sahil   |  54000 | Low         |
| Sneha   |  60000 | Medium      |
| Varun   |  57000 | Low         |
| Aditi   |  69000 | Medium      |
| Mohit   |  62000 | Medium      |
| Tanya   |  56000 | Low         |
| Dev     |  59000 | Low         |
| Nisha   |  63000 | Medium      |
| Yash    |  55000 | Low         |
| Komal   |  61000 | Medium      |
| Akash   |  67000 | Medium      |
| Divya   |  58000 | Low         |
+---------+--------+-------------+
```

7. ORDER BY-

---- SELECT EmpName, Salary
FROM Employee
ORDER BY Salary DESC;

